Chapter 3

Expectation

3.1 Expectation

Definition 3.1. Let X be a discrete random variable. The **expectation** of X, denoted by $E(X)$, is defined as:

$$E(X) = \sum_{\text{all } x} xP(X = x).$$

The expectation of X is also referred to as the **mean** of X or the **expected value** of X.

In other words, the expected value of X is a *weighted average* of the possible values that X can take on, weighted by their probabilities. If the values are of equal probability, expectation is the simple average of all x: $E(X) = \frac{1}{n}\sum x$.

The expected value of X is a *number* (if it exists), $E(X) \in \mathbb{R}$. It is not a random variable such as a function of X.

Example 3.1. The expectation of a Bernoulli random variable $X \sim \text{Bern}(p)$:

$$E(X) = 1 \times P(X = 1) + 0 \times P(X = 0) = p.$$

Example 3.2. The expectation of a Binomial random variable $X \sim \text{Bin}(n, p)$:

$$
\begin{aligned}
E(X) =& \sum_{k=0}^{n} kp(k) \\
=& \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k q^{n-k} \\
=& \sum_{k=1}^{n} n \cdot \binom{n-1}{k-1} p^k q^{n-k} \\
=& np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{n-k} \\
=& np \underbrace{\sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j}}_{\text{another Binomial PMF}} \\
=& np.
\end{aligned}
$$

Example 3.3. Life expectancy is the average number of years a person is expected to live. It is a crucial indicator of the quality of living and one of the three components of the Human Development Index (HDI) (the other two components are education and per capita GDP). Here is a toy example to compute life expectancy with hypothetical data.[1]

Age Group	No. of People	No. of Deaths	Mortality Rates	Surviving to Age	Number Surviving	Living to Exact Age
				0	1000	5
0-19	200	1	0.5%	20	995	3
20-39	300	1	0.3%	40	992	12
40-59	250	3	1.2%	60	980	78
60-79	150	12	8.0%	80	901	901
80-99	100	100	100%	100	0	0
Total	1000					

Table 3.1: Hypothetical mortality rates and life table

Note that life expectancy is **not** the average age of the population. Given the hypothetical data in the above table, if we take the midpoints of each age group

[1] This is an overly simplified example that only serves to clarify the definition of expectation. See this tutorial from MEASURE Evaluation for the actual computation of life expectancy.

and compute the weighted average,

$$\overline{\text{Age}} = (10 \times 200 + 30 \times 300 + 50 \times 250 + 70 \times 150 + 90 \times 100)/1000 = 43.$$

But this is not the number of years a person is expected to live. The computation does not even consider the chances of mortality at each age. Let X be a random variable that represents the number of years a person lives. Life expectancy should be computed as $E(X) = \sum_t tP(X = t)$.

Based on the hypothetical data, we compute the age-specific mortality rates. Suppose we have 1000 new-born babies. The chances that they would survive to age 20 is given by $1000 \times (1 - 0.5\%) = 995$. The changes that these people would survive to age 40 is $995 \times (1 - 0.3\%) \approx 992$. Similarly, we compute the rest of the surviving numbers.

To simplify the calculation, we make a bold assumption here that there is no survival between the cutoffs. For example, if a person does not survive to 40, he will die at the age 20. There is not age such as 25 or 30 in between. The number of people who die at age 20 would be those who survive the 0-19 period but does not survive the 20-39 period, which is $995 - 992 = 3$, or 0.3% of the initial population. The life expectancy is thus computed as

$$E(X) = 20 \times 0.3\% + 40 \times 1.2\% + 60 \times 7.8\% + 80 \times 90.1\% \approx 77.$$

3.2 Linearity of expectation

Theorem 3.1. *For any random variables X, Y and any constant c,*

$$E(X + Y) = E(X) + E(Y),$$
$$E(cX) = cE(X).$$

This holds regardless of the dependencies between the random variables.

Proof. The proof is not as straightforward as it seems. It is hard to combine the two random variables:

$$E(X)+E(Y) = \sum_x xP(X = x)+\sum_y yP(Y = y) \stackrel{?}{=} \sum(x+y)P(X+Y = x+y).$$

The problem becomes easier if the number of possible values of X and Y are the same and all values are equally likely,

$$E(X) + E(Y) = \frac{1}{n}\sum x + \frac{1}{n}\sum y = \frac{1}{n}\sum(x+y) = E(X+Y).$$

The original problem is equivalent to the simple case if realizing that the weighted average is jut a simple average with repetitive values. For example,

$$1 \times \frac{1}{4} + 2 \times \frac{2}{4} + 3 \times \frac{1}{4} = \frac{1}{4}(1+2+2+3).$$

Imagine the sample space as being composed of "atom" outcomes $\{\omega\}$, each with equal probability $P(\omega)$. All random variable are function of these atoms, $X(\omega)$, and $Y(\omega)$. Therefore, the expectation formula can be rewritten as

$$E(X)+E(Y) = \sum_{\omega} X(\omega)P(\omega) + \sum_{\omega} Y(\omega)P(\omega) = \sum_{\omega}(X+Y)(\omega)P(\omega) = E(X+Y).$$

Here is another way to prove linearity for discrete random variables:

$$\begin{aligned}
E(X+Y) &= \sum_{z=x+y} zP(X+Y=z) \\
E(X+Y) &= \sum_x \sum_y (x+y)P(X=x, Y=y) \\
&= \sum_x \sum_y xP(X=x, Y=y) + \sum_x \sum_y yP(X=x, Y=y) \\
&= \sum_x x \sum_y P(X=x, Y=y) + \sum_y y \sum_x P(X=x, Y=y) \\
&= \sum_x xP((X=x) \cap \bigcup_{\text{all } y}(Y=y)) + \sum_y yP(\bigcup_{\text{all } x}(X=x) \cap (Y=y)) \\
&= \sum_x xP(X=x) + \sum_y yP(Y=y) \\
&= E(X) + E(Y).
\end{aligned}$$

□

Corollary 3.1. *Further properties on the linearity of expectations:*

- If $Y = aX + b$, then $E(Y) = aE(X) + b$.

- $E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n)$
- $E(a_1X_1 + \cdots + a_nX_n + b) = a_1E(X_1) + \cdots + a_nE(X_n) + b$

Example 3.4. Redo the expectation of $X \sim \text{Bin}(n, p)$ with properties of expectation:

$$E(X) = E(X_1 + \cdots + X_n) = nE(X_i) = np$$

where $X_i \sim \text{Bern}(p)$.

Example 3.5. Let $X \sim \text{HGeom}(w, b, n)$. Find $E(X)$ the expected number of white balls. Similarly, we can decompose X:

$$X = I_1 + \cdots + I_n$$

where I_j equals 1 if the jth ball is white and 0 otherwise. We have said that $\{I_j\}$ are not independent, but the property of linearity still holds:

$$E(X) = E(I_1 + \cdots + I_n) = E(I_1) + \cdots + E(I_n).$$

Meanwhile we have

$$E(I_j) = P(j\text{-th ball is white}) = \frac{w}{w+b}$$

since unconditionally the jth ball is equally likely to be any of the balls. Thus, $E(X) = \frac{nw}{w+b}$.

Example 3.6. In a group of n people, what is the expected number of distinct birthdays among the n people (the expected number of days on which at least one of the people was born)? What is the expected number of people sharing a birthday (any day)?

Solution: Let X be the number of distinct birthdays, and write $X = I_1 + \cdots + I_{365}$, where

$$I_j = \begin{cases} 1 & \text{if someone was born on day } j \\ 0 & \text{otherwise} \end{cases}.$$

Then

$$\begin{aligned} E(I_j) &= P(\text{someone was born on day } j) \\ &= 1 - P(\text{no one was born on day } j) \\ &= 1 - \left(\frac{364}{365}\right)^n. \end{aligned}$$

Then by linearity,

$$E(X) = 365\left(1 - \left(\frac{364}{365}\right)^n\right).$$

Let Y be the number of people sharing a birthday, and $Y = J_1 + \cdots + J_n$ where J_k is an indicator that the j-th person shares his birthday with somebody else.

$$\begin{aligned} E(J_k) &= P(\text{someone shares birthday with } k) \\ &= 1 - P(\text{no one shares birthday with } k) \\ &= 1 - \left(\frac{364}{365}\right)^{n-1}. \end{aligned}$$

Therefore,

$$E(Y) = \sum_{k=1}^{n} E(J_k) = n\left(1 - \left(\frac{364}{365}\right)^{n-1}\right).$$

For some numeric values, $E(Y) = 2.3$ if $n = 30$; $E(Y) = 6.3$ if $n = 50$.

Example 3.7. Suppose that there are n people sitting in a classroom with exactly n seats. At some point, everyone got up, ran around the room, and sat back down randomly (i.e., all seating arrangements are equally likely). What is the expected value of the number of people sitting in their original seat?

Solution: Number the people from 1 to n. Let X_i be the Bernoulli random variable with value 1 if person i returns to their original seat and value 0 otherwise. Since person i is equally likely to sit back down in any of the n seats, the probability that person i returns to their original seat is $1/n$. Therefore $E[X_i] = 1/n$. Now, let X be the number of people sitting in their original seat following the rearrangement. Then $X = X_1 + X_2 + \cdots + X_n$. By linearity of expected values, we have $E[X] = \sum E[X_i] = \sum 1/n = 1$.

Example 3.8. Let Π be a permutation over $\{1, 2, \ldots, n\}$. That is a reordering of the numbers. A fixed point of a permutation are the points not moved by

the permutation. For example, in the permutation below

$$\begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \Pi & 2 & 4 & 3 & 1 \end{array}$$

The fixed point is 3. Find the expected number of fixed points of a random permutation.

Solution: Let X be the number of fixed points of a random permutation. Then $X = \sum_{k=1}^{n} \mathbf{1}_{\Pi(k)=k}$ where $\mathbf{1}_{\Pi(k)=k}$ indicates the k-th number stays the same after the permutation. By linearity,

$$E(X) = E\left(\sum_{k=1}^{n} \mathbf{1}_{\Pi(k)=k}\right) = \sum_{k=1}^{n} E\left(\mathbf{1}_{\Pi(k)=k}\right) = \sum_{k=1}^{n} \frac{1}{n} = 1.$$

Example 3.9 (Buffon's needle)**.** Rule a surface with parallel lines a distance d apart. What is the probability that a randomly dropped needle of length $l \leq d$ crosses a line?

Solution: Consider dropping any (continuous) curve of length l onto the surface. Imagine dividing up the curve into N straight line segments, each of length $\frac{l}{N}$. Let X_i be the indicator for the i-th segment crossing a line. Let X be the total number of times the curve crosses a line. Then,

$$E(X) = E(\sum X_i) = \sum E(X_i) = N \cdot E(X_i).$$

There could be infinitely many segments. It is hard to compute this expectation directly. But here we arrive an important Lemma: the expected number of crossings is proportional to the length of the curve, regardless of the shape of the curve. If we can compute $E(X)$ for some curve, the we can compute $E(X)$ for any length by scaling the value proportional to the length.

Consider a circle of diameter d. The circle always crosses the lines twice for sure. That is, $E(X_{\text{circle}}) = 2$. The length of the circle is πd. Therefore, the value of $E(X)$ for any curve of length l is given by

$$E(X) = \frac{2l}{\pi d}.$$

Now a needle can cross a line either 1 or 0 times. Thus, $E(X) = 1 \cdot P(X = 1) + 0 \cdot P(X = 0)$ is exactly the probability of a needle crossing a line.

Remark. This amazing example can be used to approximate the value of π. Let q be the probability of a needle crossing a line. q can be approximated by large number of simulations. Then $\pi \approx \frac{2l}{qd}$.

3.3 Multiplication and LOTUS

Theorem 3.2. If X and Y are independent, we have

$$E(XY) = E(X)E(Y).$$

In general, if $X_1, \ldots, X_n$ are independent, we have

$$E(X_1 X_2 \cdots X_n) = E(X_1)E(X_2)\cdots E(X_n).$$

Remark. <u>The multiplication rule will not hold without independence</u>.

Proof. For discrete and independent X, Y,

$$\begin{aligned}
E(XY) &= \sum_x \sum_y xyP(X = x, Y = y) \\
&= \sum_x \sum_y xyP(X = x)P(Y = y) \quad \text{if independent} \\
&= \sum_x xP(X = x) \sum_y yP(Y = y) \\
&= E(X)E(Y).
\end{aligned}$$

□

Remark. This is a sufficient but not necessary condition. $E(XY) = E(X)E(Y)$ does not imply independence. Consider a counter-example,

$$X = \begin{cases} 1 & (1/2) \\ 0 & (1/2) \end{cases}, \quad Z = \begin{cases} 1 & (1/2) \\ -1 & (1/2) \end{cases};$$

Then

$$Y = XZ = \begin{cases} -1 & (1/4) \\ 0 & (1/2) \\ 1 & (1/4) \end{cases}.$$

We have $E(X) = 1/2$, $E(Y) = 0$, $E(XY) = 0$. So $E(XY) = E(X)E(Y)$. But clearly X, Y are not independent.

Theorem 3.3 (Law of the unconscious statistician (LOTUS))**.** *Let X be a random variable, and g be a real-valued function of a real variable. If X has a discrete distribution, then*

$$E[g(X)] = \sum_{all\ x} g(x)P(X = x).$$

LOTUS says we can compute the expectation of $g(X)$ without knowing the PMF of $g(X)$.

Proof. The idea is similar to the one we use to prove linearity. Imagine our sample space is composed of "atoms", and $X(\omega)$ maps some atoms to numbers. The expectation of $g(X)$ can be rewritten as

$$E[g(X)] = \sum_{\omega} g(X(\omega))P(\omega).$$

If we group the atoms that compose the event $X = x$ together,

$$\begin{aligned} E[g(X)] &= \sum_{x} \sum_{\omega : X(\omega) = x} g(X(\omega))P(\omega) \\ &= \sum_{x} g(x) \sum_{\omega : X(\omega) = x} P(\omega) \\ &= \sum_{x} g(x)P(X = x). \end{aligned}$$

□

Example 3.10. Compute $E(X)$ and $E(X^2)$ given the following distribution.

X	0	1	2
X^2	0	1	4
P	1/4	1/2	1/4

Solution: According to the distribution table, we compute the expectations as

$$\begin{aligned} E(X) &= 0 \times 1/4 + 1 \times 1/2 + 2 \times 1/4 = 1; \\ E(X^2) &= 0 \times 1/4 + 1 \times 1/2 + 4 \times 1/4 = 3/2. \end{aligned}$$

Note that $E(X^2) \neq [E(X)]^2$.

Remark. In general, $E[g(X)] \neq g(E(X))$. Linearity implies that if g is a linear function of X, then $E[g(X)] = g(E(X))$. For a nonlinear function g, the relationship between $E[g(X)]$ and $g(E(X))$ is determined case by case. We will get back to this point when we learn Jensen's inequality.

Example 3.11 (St. Petersburg Paradox)**.** Flip a fair coin over and over again until the head lands the first time. You will win 2^k dollars if the head lands in the k-th trial (including the successful trial). What is the expected payoff of this game?

Solution: Let $X = 2^k$. We want to find $E(X)$. The probability of the first head showing up in the k-th trial is $\frac{1}{2^k}$. Therefore,

$$E(X) = \sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k} = \sum_{k=1}^{\infty} 1 = \infty$$

The expected payoff is infinitely high! This is against most people's intuition. This is because we intuitively think that $E(X) = E(2^k) = 2^{E(k)}$, which is a finite number.